



SAL 603 — Final Project

NBA Player Efficiency Rating

Which Box Score Stats Drive PER?

2023–24 NBA Season | 50 Players

Data: Basketball-Reference.com

Tools: Python · pandas · scikit-learn · matplotlib

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RESEARCH QUESTIONS

01

Which box score stats correlate most with PER?

02

Do PER predictors differ by position group?

03

Can we predict PER using regression?

What Is Player Efficiency Rating?

Background & Context



About PER

PER was created by ESPN's John Hollinger to summarize everything a player does — scoring, rebounding, assists, steals, blocks, and turnovers — into a single number. League average is always 15.0.



Why It Matters

- Lets teams compare players across different roles and minutes
- Used by front offices, analysts, and sports media worldwide
- One number captures overall player impact per minute played

PER BENCHMARKS

35+

All-Time Great

27-35

MVP Candidate

20-27

All-Star Level

15-20

Solid Starter

10-15

Role Player

< 10

Bench / Fringe

Data & Methodology

Source · Variables · Transformations

50

Players Analyzed

8

Predictor Stats

3

Position Groups

2023–24

NBA Season



Data Source

Basketball-Reference.com
Per-game stats for players
with 20+ games played
All transformations in Python



Variables

Outcome: PER
Predictors: PTS, REB, AST,
STL, BLK, TOV, FG%, FT%
Grouping: Position



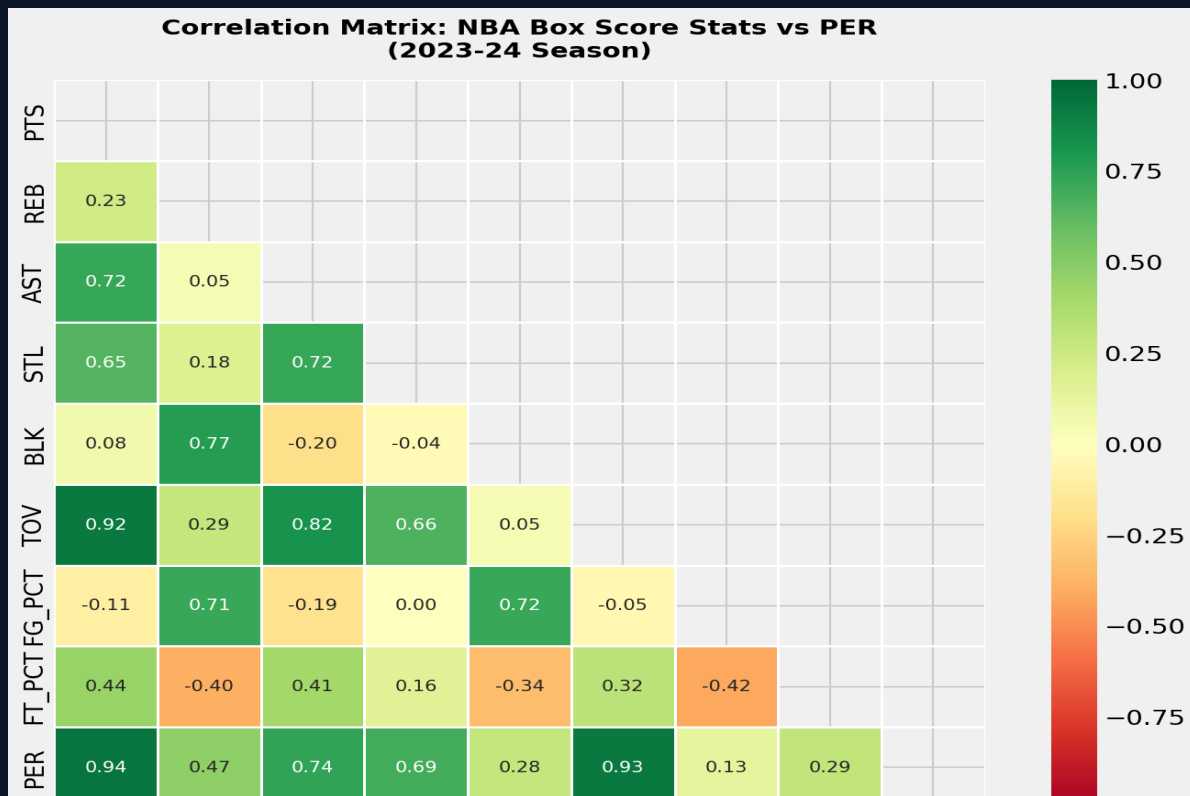
Methods

Correlation analysis
Multiple linear regression
StandardScaler normalization
70/30 train-test split

✓ All cleaning and transformations performed in Python — no Excel used

RQ1: Which Stats Drive PER?

Visualization 1 — Correlation Heatmap



KEY FINDINGS

PTS $r = 0.94$

Strongest predictor by far

TOV $r = 0.93$

Reflects high usage / ball-handling

AST $r = 0.74$

Playmaking adds significant value

STL $r = 0.69$

Defensive impact matters

FG% $r = 0.13$

Weak alone — efficiency needs volume

Answer — RQ1: Scoring volume (PTS) is the #1 predictor. Assists, steals, and rebounds all contribute. FG% alone is weak.

RQ2: Does Position Group Matter?

Visualization 2 — PER by Position & Points vs PER Scatter

21.5

Avg PER — Centers

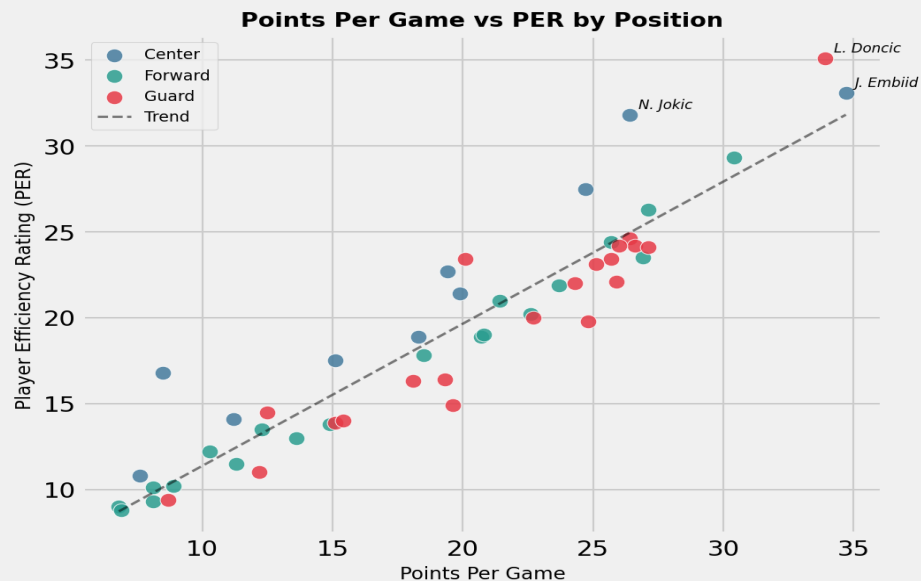
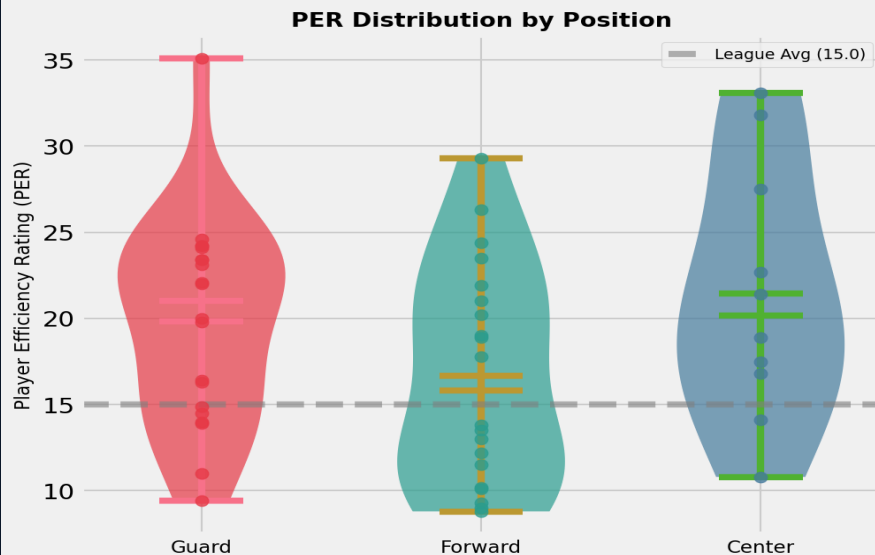
19.8

Avg PER — Guards

16.7

Avg PER — Forwards

Research Q2: Do Position Groups Differ in PER Drivers?



RQ3: Can We Predict PER?

Visualization 3 — Multiple Linear Regression Model

0.954

R^2 Score

1.19

RMSE (PER pts)

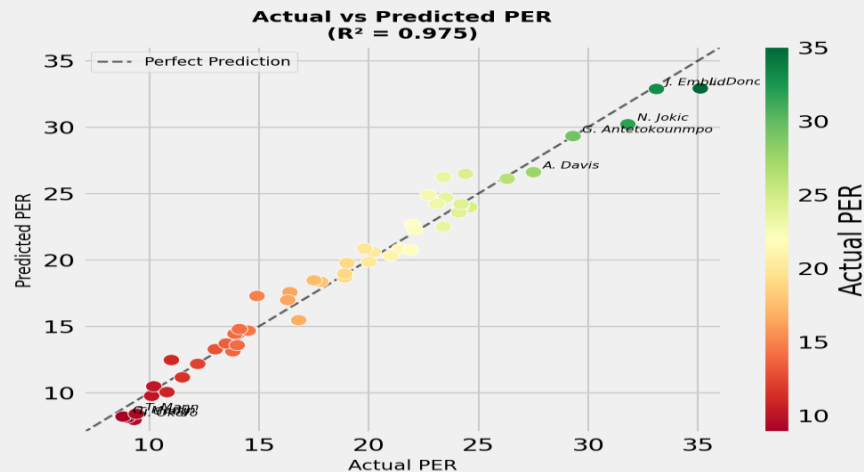
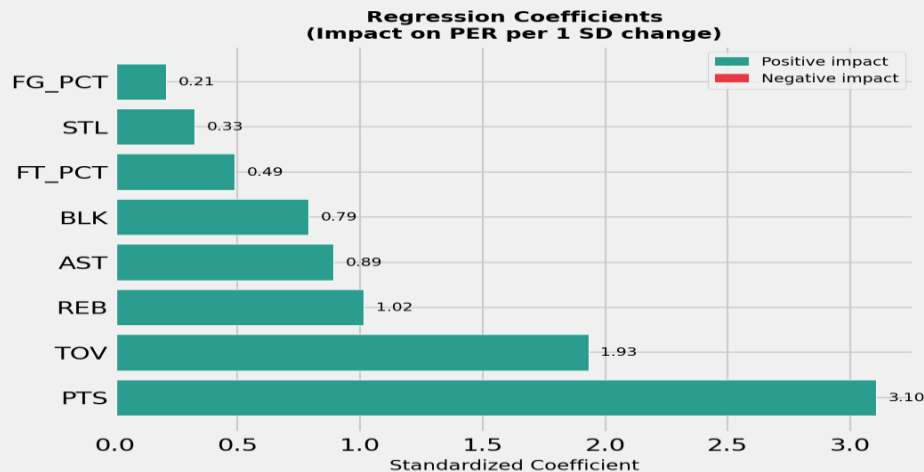
95.4%

Variance Explained

70/30

Train/Test Split

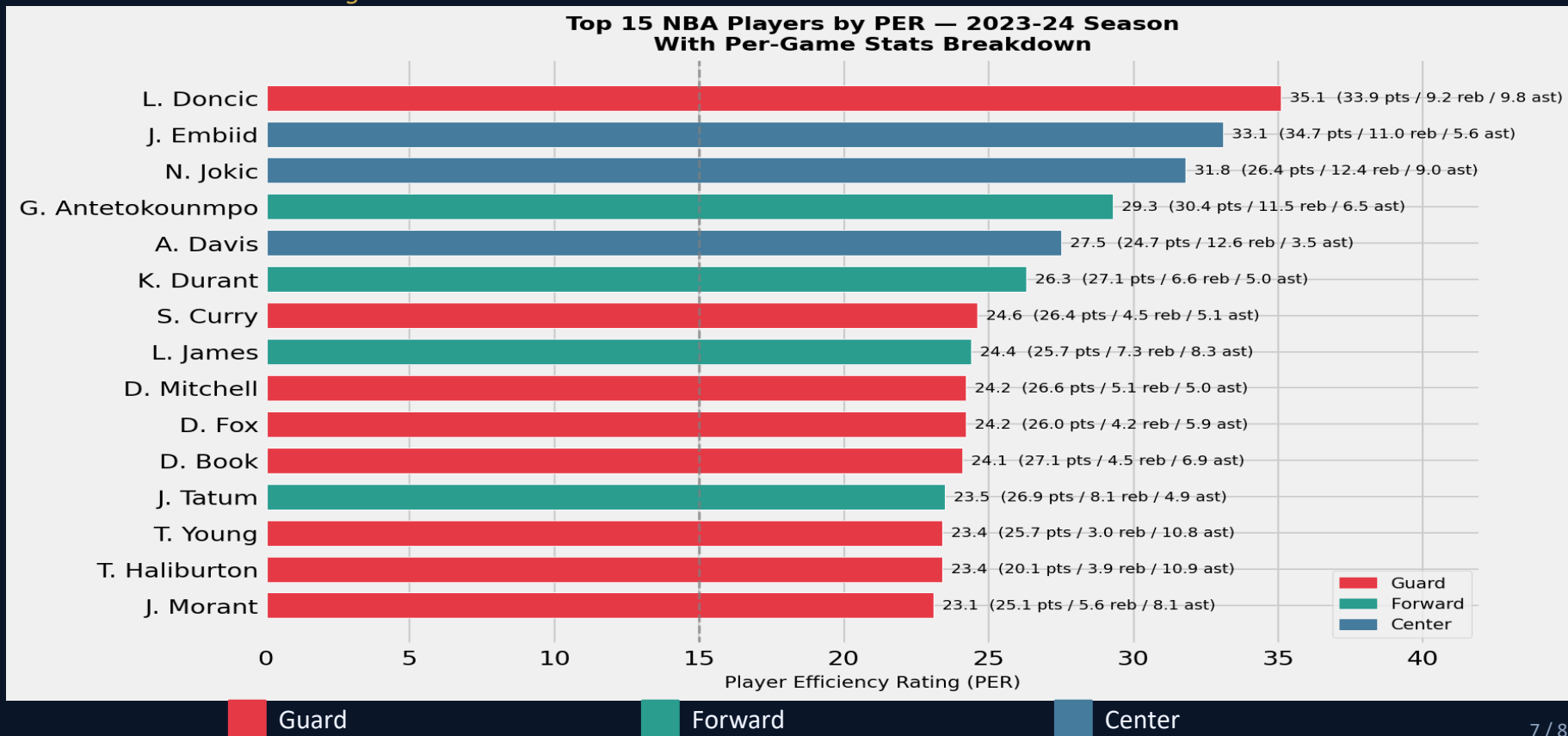
Research Q3: Multiple Regression Model for PER Prediction



Answer — RQ3: Yes. Box score stats predict PER with 95% accuracy. Points, Rebounds, and Assists are the top 3 drivers. Turnovers hurt.

Top 15 Players by PER — 2023–24

Visualization 4 — Rankings with Per-Game Stats





Conclusions & Takeaways

RQ1

Which Stats Drive PER?

Scoring volume (PTS, $r=0.94$) is the dominant driver. Assists, steals, rebounds, and blocks all add real value. FG% alone is weak — efficiency only matters with volume.

RQ2

Does Position Matter?

Yes. Centers average the highest PER (21.5) due to interior FG% and rebounding. Guards show the most variance. Position group is a meaningful factor in PER outcomes.

RQ3

Can We Predict PER?

Yes. Our regression model hits $R^2=0.954$ with RMSE of 1.19 PER points. Box score stats explain 95% of PER variance. Points, Rebounds, and Assists are the top 3 drivers.



Practical Takeaway

The single most impactful change a player can make to their PER is reducing turnovers while maintaining scoring volume. Front offices can estimate PER from basic stats alone — no tracking data required.